UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 001 · Introduction to second and higher order DE with constant coefficients · compiled 15 October 2026

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1 Short notes

1 · Linear differential equations and their order

A DE is linear when y and its derivatives occur only in the first degree and are not multiplied together; coefficients may be functions of x or constants, never of y. Order = highest derivative. Linear of order n: $y^{(n)} + a_1 y^{(n-1)} + \cdots + a_n y = X$. Example: $\frac{dy}{dx} + 2xy = x^2$ is linear; $y^2 \frac{dy}{dx} + \sin y = x^2 e^{2y}$ is not.

2 · Constant or variable coefficients; homogeneous or not

Two forms: constant coefficients ($2y''' - 5y'' + 7y' + y = x^2$) and variable coefficients ($2x^3 y''' - (5x^2 + 2)y'' + xy' + y = x^2 \sin x$). Homogeneous: $X = 0$ ($2y'' - 3y' + 9y = 0$); non-homogeneous: X a nonzero function of x ($2y'' - 3y' + 9y = x \sinh x$). Putting $X = 0$ gives the associated homogeneous equation.

3 · The differential operator D and $f(D)$

$D = \frac{d}{dx}$, $D^n = \frac{d^n}{dx^n}$. The LDE becomes $f(D)y = X$ with $f(D) = D^n + a_1 D^{n-1} + \cdots + a_n$, the linear differential operator. With constant coefficients $f(D)$ is a polynomial in D: factorise it, in any order. Preview: $f(D)e^{mx} = f(m)e^{mx}$ (class 002).

4 · Linear independence and the complementary function

$f_1, \ldots, f_n$ are linearly dependent if $c_1 f_1 + \cdots + c_n f_n = 0$ with constants not all zero; otherwise independent. Two functions: independent if neither is a constant multiple of the other (e^x, e^{2x} yes; $e^x, 3e^x$ no). If $y_1, \ldots, y_n$ are n independent solutions of the homogeneous LDE, the complete solution $c_1 y_1 + \cdots + c_n y_n$ is the complementary function.

5 · The general solution: $y = \text{CF} + \text{PI}$

For $f(D)y = X$: $y = y_c + y_p$, where y_c (CF) is the complete solution of $f(D)y = 0$ and y_p (PI) is one particular solution of $f(D)y = X$. CF: n constants; PI: none. Linearity gives $f(D)(y_c + y_p) = X$, and the PI of a sum of terms is the sum of their PIs. Every Unit 2 ESA answer has this form.

2 Keyword index

21 terms; the number in brackets is the topic.

Associated homogeneous equation (2). $f(D)y = 0$ for $f(D)y = X$.

Auxiliary equation (3). $f(m) = 0$ (class 002).

Complementary function (4). $y_c = c_1y_1 + \cdots + c_ny_n$, complete solution of $f(D)y = 0$.

Complete solution (5). $y = y_c + y_p$.

Constant coefficients (2). All a_i numbers.

Differential operator (3). $D = \frac{d}{dx}$.

Higher-order LDE (1). Linear, order 2 or more.

Homogeneous LDE (2). RHS $X = 0$.

Linear DE (1). y and derivatives first degree, not multiplied together.

Linear differential operator (3). $f(D) = D^n + a_1D^{n-1} + \cdots + a_n$.

Linearly dependent (4). $\sum c_if_i = 0$ with c_i not all zero.

Linearly independent (4). $\sum c_if_i = 0$ only for all $c_i = 0$.

Non-homogeneous LDE (2). RHS X a nonzero function of x.

Non-linear DE (1). Not linear: yy', y^2, $\sin y$ ...

Operator form (3). $f(D)y = X$.

Order (1). Highest derivative present.

Ordinary DE (1). One independent variable; linear or non-linear.

Particular integral (5). y_p: one solution of $f(D)y = X$, no constants.

Superposition (4). Combinations of solutions of a homogeneous LDE are solutions.

Variable coefficients (2). Some a_i functions of x.

Wronskian (4). $y_1y_2' - y_2y_1'$; nonzero means independent.

3 Rapid recall

1	When is a DE linear?	y and its derivatives occur in the first degree and are not multiplied together.
2	Is $y^2y' + \sin y = x^2e^{2y}$ linear?	No.
3	Order of a DE?	Order of the highest derivative present.
4	General linear DE of order n?	$y^{(n)} + a_1y^{(n-1)} + \cdots + a_ny = X$.
5	Constant against variable coefficients?	a_i numbers against a_i functions of x.
6	Homogeneous LDE?	RHS $X = 0$.
7	Is $2x^3y''' - (5x^2+2)y'' + xy' + y = x^2\sin x$ linear?	Yes: order 3, variable coefficients, non-homogeneous.
8	D and D^2 mean?	$\frac{d}{dx}$ and $\frac{d^2}{dx^2}$.
9	$f(D)$ for $y'' - 5y' + 6y$?	$D^2 - 5D + 6 = (D-2)(D-3)$.
10	Do the factors of $f(D)$ commute?	Yes, with constant coefficients only.
11	y_1, y_2 independent when?	Neither is a constant multiple of the other.
12	Are e^x and $3e^x$ independent?	No: ratio 3.
13	$W(e^x, e^{2x})$?	e^{3x}.
14	CF of an n-th order homogeneous LDE?	$c_1y_1 + \cdots + c_ny_n$ with y_i independent solutions.
15	Complete solution of $f(D)y = X$?	$y = y_c + y_p$ = CF + PI.
16	How many constants in the PI?	None; all n are in the CF.
17	PI of $(D^2 - 3D + 2)y = e^{3x}$?	$\frac{1}{2}e^{3x}$.
18	PI of $X_1 + X_2$?	PI of X_1 plus PI of X_2.
19	How many Unit 2 ESA occurrences use CF + PI?	All 24 (19 distinct questions).

Mnemonic (study aid). *"Constants live in the CF; the PI travels alone."* The PI carries no arbitrary constants (study aid).

Mnemonic (study aid). *"Ratio constant, dependent."* The quick test for two functions (study aid).

4 Last 10 minutes before the exam

Top facts

1. Linear: y and derivatives first degree, not multiplied; coefficients functions of x or constants.
2. Order n LDE: $y^{(n)} + a_1 y^{(n-1)} + \cdots + a_n y = X$; constant or variable coefficients.
3. Homogeneous: $X = 0$; non-homogeneous: X a nonzero function of x.
4. $D = \frac{d}{dx}$; the LDE is $f(D)y = X$, $f(D) = D^n + a_1 D^{n-1} + \cdots + a_n$; factors commute (constant coefficients).
5. Independent: $\sum c_i f_i = 0$ only for all $c_i = 0$; two functions: ratio not constant.
6. CF $= c_1 y_1 + \cdots + c_n y_n$ (y_i independent solutions of $f(D)y = 0$): n constants.
7. Complete solution $y = y_c + y_p$ = CF + PI; the PI has no constants.
8. All 24 Unit 2 ESA occurrences are answered as CF + PI.

Top formulas

E1.3 · Linear DE of order n — ESA ★★

$$\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \cdots + a_n y = X$$

GIVES The general linear equation of order n.

APPLIES $a_1, \ldots, a_n$ functions of x or constants; X a function of x alone.

NOTE Slide 001·7 writes $P_1, \ldots, P_n$; slide 001·8 onward $a_1, \ldots, a_n$.

S001·7 · S001·8 · T001 00:08:38 · T001 00:09:46

E3.2 · The equation as $f(D)y = X$ — ESA ★★★

$$f(D)\, y = X, \qquad f(D) = D^n + a_1 D^{n-1} + a_2 D^{n-2} + \cdots + a_n$$

GIVES The compact form used in every later class.

APPLIES Constant a_i: $f(D)$ is a polynomial in D.

NOTE a_n multiplies y itself ($D^0 = 1$).

S001·11 · S001·12 · T001 00:16:29

E4.4 · Superposition: the complementary function — ESA ★★★

$$y_c = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n$$

GIVES The complete solution of the homogeneous LDE $f(D)y = 0$ of order n.

APPLIES $y_1, \ldots, y_n$ are n linearly independent solutions.

NOTE Second order: $y_c = c_1 y_1 + c_2 y_2$. Exactly n arbitrary constants.

S001·15 · S001·16 · T001 00:24:29 · T001 00:25:14

E5.1 · Complete solution of $f(D)y = X$ ESA ★★★

$$y = y_c + y_p = \text{CF} + \text{PI}$$

GIVES The answer to every non-homogeneous LDE.

APPLIES y_c: complete solution of $f(D)y = 0$; y_p: any one solution of $f(D)y = X$.

NOTE All 24 Unit 2 ESA occurrences are answered in this form.

S001·12 · S001·17 · T001 00:27:16 · T001 00:27:34

E5.2 · Why it works (linearity) ESA ★★

$$f(D)(y_c + y_p) = f(D)y_c + f(D)y_p = 0 + X = X$$

GIVES $y_c + y_p$ solves $f(D)y = X$ and has all n constants.

APPLIES $f(D)$ linear.

NOTE Also: the PI of $X_1 + X_2$ is the sum of the PIs of X_1 and X_2 (W5.3).

ADDED FOR CLARITY S001·17